HiGreen: Hierarchical Reinforcement Learning for Carbon-Aware Geo-Distributed Datacentre Scheduling with Renewable Look Ahead

Mingwei Li, Chuanru Ren, John Panneerselvam, Mengqi Li, and Lu Liu

***Abstract*—The rapid growth of computationally intensive AI workloads significantly exacerbates datacentre electricity demands. While datacentres increasingly adopt renewable energy, its intermittent nature often forces them to fall back on carbon-intensive grid power. To address this mismatch, we propose *HiGreen*, a hierarchical Deep Reinforcement Learning (DRL) framework designed to proactively align computing workloads with fluctuating green energy supplies across geographically distributed datacentres. To achieve this proactive scheduling, the system must anticipate future energy availability. However, injecting raw, complex forecasting data directly into an AI scheduler often overwhelms the learning process and causes severe training instability. To overcome this, our core innovation is a dual-temporal architecture that simplifies how the system processes the future insights. Instead of using raw prediction arrays, HiGreen employs a pre-trained forecasting backbone to compress volatile green energy dynamics into simple, high-level indicators, such as upcoming peaks and trends. A global routing agent then combines these forward-looking indicators with its own historical memory to make carbon-efficient workload distribution decisions. Simultaneously, local schedulers independently manage the physical placement of tasks within each datacentre to ensure performance constraints are met. Experimental evaluations across diverse workload scales (5k–100k cloudlets) demonstrate that HiGreen outperforms traditional baselines, consistently achieving up to a 3.08% reduction in total carbon emissions and an 11.69% reduction in carbon intensity without compromising service quality.**

***Index Terms*—Carbon-aware Scheduling, Hierarchical Reinforcement Learning, Geo-distributed Cloud Systems, Decision-aware Renewable Integration**

I. Introduction

Information and Communication Technology (ICT), particularly datacentres underpin modern cloud services and latency-sensitive applications. However, the rapid growth of computational demand driven by AI and LLM services has precipitated a severe global energy crisis [1]. ICT infrastructure is projected to consume up to 20% of global electricity by 2030 [2]. For example, leading datacentres like Google and Microsoft have consumed 30.8 TWh (Terawatt-hour) and 24 TWh respectively in 2024, and emitted 11.5 million metric tons and 25.3 million metric tons of CO_2 respectively [3], [4].

M. Li, J. Panneerselvam, M. Li, and L. Liu are with the Department of Computer Science, University of Exeter, Exeter EX4 4QF, U.K. (e-mail: {ml969, j.panneerselvam, ml1065, l.liu3}@exeter.ac.uk).

C. Ren is with the School of Computer Science and Technology, Tongji University, Shanghai 201804, China (e-mail: rencr@tongji.edu.cn).

Corresponding author: Lu Liu.

Initial efforts to mitigate the soaring energy consumption and carbon emissions focused on hardware-centric optimisation within individual datacentres, such as Dynamic Voltage and Frequency Scaling (DFVS) and intensive cooling management [5], [6]. However, these physical optimisations are rapidly approaching their theoretical limits since their PUE is approaching 1.0. Moreover, they only reduce total power consumption without changing the underlying carbon intensity of the energy supply. A datacentre operating on a fossil-fuel grid will still consume brown energy.

Beyond hardware tuning, software-level techniques within a single datacentre—such as workload consolidation, VM live migration, and dynamic server shutdown—can further reduce energy waste by improving resource utilisation [7]. While effective at minimising idle power, these intra-datacentre strategies remain constrained by the carbon intensity of the local grid: consolidating workloads onto fewer servers in a coal-powered region still yields high-carbon computation. Therefore, meaningful carbon reduction requires looking *beyond* the boundaries of a single datacentre.

This limitation has naturally driven research towards geo-distributed datacentres integrated with renewable energy sources (RES). The premise is appealing: by distributing computation across geographically distributed sites, operators can exploit regional differences in renewable availability and grid carbon intensity. However, fully realising this premise faces the obstacles. First, RES are inherently intermittent and volatile; solar output fluctuates with cloud cover and times, while wind power varies with weather patterns, creating sharp mismatches between energy supply and computational demand. Second, long-distance energy transmission incurs non-trivial losses—annual electricity transmission and distribution losses average approximately 5% in modern grids and are substantially higher in regions with ageing infrastructure [8]—making it inefficient to transport surplus renewable energy from one region to serve computation in another. Third, while battery storage could in principle buffer the intermittency, deploying utility-scale storage at the capacity required by hyperscale datacentres remains prohibitively expensive. Recent industry analyses indicate that lithium-ion battery costs for datacentre-scale deployments (tens to hundreds of MWh) still range between \$200–\$400/kWh for installed systems [9], and the associated space, cooling, and maintenance overheads further undermine economic viability. As a result, massive amounts of renewable energy are frequently wasted globally due to inadequate storage infrastructure and transmission bottlenecks. This exposes a severe *spatiotemporal mismatch* between highly intermittent energy production and computational demand.

A fundamental paradigm shift is in need. Our vision is to lead a paradigm shift in ICT sustainability, driven by responsible AI integration, to reduce large-scale ICT carbon footprints. While mission-critical or ultra-low-latency applications (e.g., high-frequency financial trading, real-time control systems) strictly demand data locality and cannot be arbitrarily deferred, a massive and rapidly growing proportion of energy-intensive cloud workloads—such as AI model training, large-scale batch processing, and scientific simulations—exhibits significant spatial and temporal flexibility. Unlike traditional strategies that attempt to transmit energy across long distances to meet static compute demands, we propose a novel paradigm that proactively routes cloud workloads to datacentres currently experiencing green energy surpluses. This approach shifts the operational focus from grid-level energy transmission to intelligent workload scheduling, thereby achieving an end-to-end bilateral alignment between datacentre computing demands and renewable energy production. This alignment is driven by high-precision predictions of renewable availability, allowing the system to proactively dispatch cloud tasks to the most carbon-efficient locations.

In practice, transiting to this novel paradigm remains challenging. The difficulties lie in reconciling the high-dimensional, non-stationary nature of the green energy dynamics with strict, low-latency feasibility constraints of local Virtual Machine (VM) placement. Existing DRL approaches [10]–[12] typically treat scheduling as an instantaneous, reactive decision based on current states of datacentres. Because they lack explicit future foresight regarding renewable non-stationarity and fail to decouple cross-datacentre coordination from local execution, these single-layer models struggle to proactively route workloads towards datacentres with forthcoming renewable surplus.

Addressing the inherent spatiotemporal mismatch between long-term renewable intermittences and instantaneous local execution constraints necessitates a decoupled, forecast-driven decision-making paradigm. To address traditional reactive limitations, the global scheduler must proactively anticipate future renewable dynamics. However, directly integrating raw forecasting trajectories into a reinforcement learning state space poses severe scientific challenges. It forces the agent to simultaneously learn feature extraction and action selection, inevitably inducing representational entanglement. Furthermore, it exposes the value function to the unmitigated propagation of epistemic uncertainty and high-frequency forecasting noise. To overcome these theoretical bottlenecks, this paper proposes HiGreen, a hierarchical DRL framework driven by an *explicit-implicit dual temporal modelling* architecture. Rather than relying on raw forecasting data, HiGreen employs *semantic state compression* to distil intermittent future dynamics into multi-scale structural priors using a pre-trained multivariate forecasting backbone (TimeCAP) [13]. By integrating these explicit future priors with implicit historical states encoded by a GTrXL global routing policy, the framework effectively bridges the gap between proactive, cross-datacentre load balancing and reactive, feasibility-constrained local scheduling. Experiments in an end-to-end simulator-to-RL pipeline demonstrate consistent reductions in brown-energy consumption and carbon emissions across diverse workload scales.

The following are the main contributions of this work:

- Formulate the geo-distributed carbon-aware scheduling problem as a foresight-driven hierarchical decision process. This formulation explicitly addresses the *spatiotemporal mismatch* between volatile green energy and strict QoS constraints, shifting the paradigm from reactive optimisation to proactive, bilateral supply-demand alignment.
- Propose a *spatiotemporally decoupled* hierarchical DRL framework to resolve the conflict between long-term global carbon efficiency and instantaneous local execution. By decoupling global workload routing from local VM placement, we ensure scalable coordination without violating strict physical boundaries.
- Design an *explicit-implicit dual temporal modelling* mechanism integrating the TimeCAP forecasting backbone with a GTrXL policy. By employing *semantic state compression* to extract structural priors, we effectively circumvent the representational entanglement and noise propagation inherent in raw predictive DRL.
- Comprehensively evaluate HiGreen across diverse dynamic workloads (5k–100k cloudlets). Empirical results demonstrate up to a 3.08% reduction in total carbon emissions and an 11.69% reduction in carbon intensity without compromising strict QoS guarantees, significantly outperforming representative baselines.

The remainder of this paper is organised as follows. Section II reviews related works on renewable-aware and carbon-aware scheduling. Section III introduces the system model and formulates the hierarchical scheduling problem. Section IV presents HiGreen, including the renewable look-ahead module, state/action design, reward formulation, and policy architecture. Section V describes the experimental setup and baselines, and reports and analyses the results. Finally, Section VI concludes the paper and outlines future work.

II. Related Work

This section reviews existing works on renewable-aware scheduling for cloud datacentres, with an emphasis on geo-distributed workload orchestration and carbon-aware objectives. Existing approaches can be categorised into: (i) mathematical optimisation, (ii) heuristics and rule-based controllers, (iii) meta-heuristics, and (iv) learning-based (AI/DRL) scheduling. We summarise representative techniques and highlight their research gaps.

A. Mathematical Optimisation

Mathematical optimisation (e.g., MILP/MINLP) provides principled formulations for jointly optimising cost, energy, and QoS constraints in datacentres. For example, queueing-aware formulations with DVFS-enabled servers can yield near-optimal operating points for hybrid-powered geo-distributed datacentres, but the computational cost often limits their real-time applicability at scale [14]. MILP-based methods are used to design and operate hybrid renewable systems for datacentres, achieving deterministic solutions under strong assumptions on forecast accuracy and system parameters [15]. Furthermore, optimisation-based energy management in fully renewable multi-energy systems offers detailed modelling, yet often focuses on day-ahead planning and remains sensitive to uncertainty and modelling complexity [16], [17]. Overall, mathematical optimisation methods provide strong performance baselines but frequently face severe scalability and adaptivity challenges in large stochastic environments.

B. Heuristics and Rule-based Scheduling

Traditional cloud schedulers often relied on static policies (e.g., Round Robin and Shortest Job First), which are simple and incur low overheads. However, such methods typically lack adaptivity under dynamic workloads and heterogeneous resources [18]. Extensions incorporating energy awareness commonly applied hand-crafted rules, for instance, DVFS-based heuristics that adjust processor frequency and voltage to reduce power while meeting task constraints [19].

To handle spatial heterogeneity, computational shifting has been studied as an alternative approach to physical storage and long-distance energy transmission. For example, migrating workloads towards renewable-rich regions (sometimes described as a "virtual battery") can significantly improve local renewable utilisation [20]. Moreover, topology-aware heuristics, such as joint server–network scheduling, can further reduce communication costs and network energy [21]. However, frequent migration induces non-trivial network overhead, and these heuristics typically require strong assumptions on workload structures, making them less efficient in generalising across diverse cloud traces.

Hierarchical control structures have also been explored, combining a coarse-grained global controller with fine-grained local controllers for VM/host-level management [7]. While conceptually aligned with hierarchical scheduling, rule-based frameworks are fundamentally limited by their fixed decision logic. They operate reactively, failing to naturally integrate predictive signals to proactively anticipate renewable fluctuations or holistically optimise long-term carbon efficiency across distributed geographies.

C. Meta-heuristics

Meta-heuristics (e.g., GA/PSO/ACO and hybrids) are widely adopted for multi-objective scheduling when problems are NP-hard with complex constraints [22]–[24]. Such approaches are applied to achieve workflow scheduling across geo-distributed datacentres and energy-aware scheduling in fog/cloud settings, often combining global search (exploration) with local refinement (exploitation) [25]–[29]. While these methods can obtain high-quality solutions, they typically require careful parameter tuning and incur high computational overheads, lack ubiquitous deployment in production systems, and do not naturally support continual online learning under streaming workload arrivals and non-stationary renewable supply.

D. Learning-based Scheduling and Forecast-aware Control

Learning-based approaches, especially DRL, have gained traction for sequential scheduling under uncertainty [30]. Recent systems combine prediction and DRL controllers to improve renewable utilisation and reduce brown energy consumption while maintaining QoS [31], [32]. Another work integrates RL with optimisation (e.g., ILP/MPC) and introduces uncertainty-aware forecasting to improve robustness against renewable variability [33].

For geo-distributed settings, RL-based workload shifting is explored to exploit spatial diversity in renewable availability and electricity carbon intensity [34]–[37]. DRL-based schedulers such as GreenDRL and RARE demonstrate reductions in energy utilisation and carbon emission under appropriate reward design and observability assumptions in singular datacentre environment [10], [11]. Multi-agent reinforcement learning (MARL) further enables decentralised control and can address partial observability via centralised training and decentralised execution (CTDE), for example by coupling global routing decisions with local placement decisions [38].

A recurring limitation in the existing literature is the struggle to reconcile the geographical distribution of renewable energy with immediate, node-level execution constraints. While recent DRL schedulers increasingly incorporate forecasting to achieve look-ahead capabilities, directly feeding unprocessed, granular prediction trajectories into the learning agents forces them to simultaneously act as forecasters and schedulers. This inevitably induces severe representational entanglement and propagates high-frequency forecasting noise, fundamentally destabilising the learning of robust, forward-looking routing strategies. To bridge this gap, HiGreen advances beyond simplistic forecast-appended methods via a multi-tiered framework that distils future renewable dynamics into compact structural priors using the TimeCAP semantic state compression engine. By embedding this compressed predictive intelligence rather than raw trajectories, our approach circumvents theoretical DRL bottlenecks, equipping the global scheduling agent with proactive foresight while preserving a highly efficient, low-overhead state representation, ultimately achieving system-wide carbon reduction without compromising local operational boundaries.

III. System Model and Problem Formulation

A. Problem Definition

We consider a geo-distributed cloud environment composed of multiple heterogeneous datacentres located in different regions. Each datacentre operates under time-varying renewable energy availability and region-dependent carbon factors. Cloud workloads arrive dynamically over time, with their resource demands known upon arrival.

The system follows a green-first energy usage principle: renewable energy is utilised whenever available, while residual demand is supplied by the power grid. Excess renewable generation that cannot be immediately consumed is curtailed and considered as wasted, and no energy storage is assumed. At runtime, the operational state of each datacentre, including resource utilisation and queue backlog, is observable.

Task scheduling is performed through a hierarchical decision process. At the global level, workloads arriving are routed across distributed datacentres for execution. At the local level, tasks assigned to a data centre are allocated to appropriate virtual machines based on instantaneous resource availability.

The objective is to minimise long-term carbon emissions and renewable energy wastage while maintaining acceptable QoS, such as bounded waiting time and queue backlog. This scheduling problem is inherently intractable due to a fundamental spatiotemporal conflict: the spatial heterogeneity of carbon intensities must be continuously reconciled with the temporal non-stationarity of intermittent renewable supplies. Classical reactive scheduling formulations fail to capture the long-term temporal dependencies required for carbon efficiency. Conversely, directly integrating raw forecasting trajectories into a unified global state space forces the agent to simultaneously learn feature extraction and action selection. This inevitably induces severe representational entanglement and exposes the policy to high-frequency forecasting noise, rendering traditional single-layered learning approaches fundamentally unstable.

To overcome this, we redefine the geo-distributed scheduling problem as a spatiotemporally decoupled hierarchical Markov Decision Process. This formulation strictly isolates the long-horizon, predictive global routing objectives from the short-horizon, strict-latency local physical constraints, thereby ensuring mathematical tractability and operational scalability.

B. System Model

Fig. 1 illustrates the high-level architecture of our hierarchical scheduling system for geo-distributed datacentres. We consider N datacentres that are heterogeneous in their infrastructure configurations (host profiles, VM fleets, and region-dependent carbon factors). Each datacentre is associated with a renewable-power trace (wind/solar) that is aligned to its geographic region considering time-zone offsets. At runtime, each datacentre draws IT power from its host power models; renewable energy is used first and any remaining demand is supplied by grid electricity, while excess renewable generation is dropped (no storage).

The system follows a two-level hierarchy. At the *global level*, a global scheduler performs inter-datacentre routing: arriving cloud workloads are appended to a global waiting queue, and the scheduler periodically selects a batch of tasks from this queue and assigns each task to a target datacentre. At the *local level*, each datacentre is managed by a local broker and a local scheduler, which maintains a FIFO waiting queue and dispatches tasks to feasible VMs within the same datacentre.

Both the global scheduler and the local schedulers are instantiated as deep reinforcement learning agents. During inference, the global routing agent consumes a dual-temporal aggregated observation. This uniquely fuses the implicit operational metrics of all datacentres with the explicit multi-scale semantic priors distilled by the forecasting backbone, outputting proactive routing actions for the current batch. After routing, workloads are transferred to the corresponding local waiting queues, where each local agent repeatedly selects tasks from its FIFO queue and assigns them to VMs based strictly on instantaneous host/VM utilisation and physical resource availability.

C. Energy and Carbon Emission Model

This subsection specifies the way datacentre-level energy consumption and carbon emissions are computed in the simulator.

Host utilisation and IT power. Let $u_{i,h}(t) \in [0, 1]$ denote the CPU utilisation of host $h \in \mathcal{H}_i$ in datacentre DC_i at simulation time t. The instantaneous IT power of host h is obtained from its utilisation-driven power model:

$$p_{i,h}(t) = P_{i,h}(u_{i,h}(t)) \,. \tag{1}$$

The datacentre IT power is the sum of all hosts:

$$p_i(t) = \sum_{h \in \mathcal{H}_i} p_{i,h}(t). \tag{2}$$

Step energy accounting. Consider a simulation step with elapsed wall-clock time Δt (hours) between two consecutive updates. The IT energy demand of DC_i during this step is

$$\Delta E_i^{\text{dem}}(t) = p_i(t)\,\Delta t \qquad (\text{Wh}). \tag{3}$$

Renewable supply and green-first allocation (no storage). Let $g_i(t)$ denote the instantaneous renewable power available at DC_i (W), aggregated from renewable traces (wind/solar) at time t. The available renewable energy within the step is

$$\Delta E_i^{\text{avail}}(t) = g_i(t)\,\Delta t \qquad (\text{Wh}). \tag{4}$$

We assume a green-first policy with no storage:

$$\Delta E_i^{\text{green}}(t) = \min\left(\Delta E_i^{\text{dem}}(t),\, \Delta E_i^{\text{avail}}(t)\right), \tag{5}$$

$$\Delta E_i^{\text{brown}}(t) = \Delta E_i^{\text{dem}}(t) - \Delta E_i^{\text{green}}(t), \tag{6}$$

$$\Delta E_i^{\text{waste}}(t) = \Delta E_i^{\text{avail}}(t) - \Delta E_i^{\text{green}}(t). \tag{7}$$

Carbon emissions and carbon intensity. Each datacentre DC_i is associated with region-dependent carbon factors η_i^{brown} and η_i^{green} (kgCO_2/kWh) for grid electricity and renewable energy respectively. Carbon emissions are computed as

$$\Delta C_i(t) = \eta_i^{\text{green}} \frac{\Delta E_i^{\text{green}}(t)}{1000} + \eta_i^{\text{brown}} \frac{\Delta E_i^{\text{brown}}(t)}{1000}, \tag{8}$$

where $\Delta C_i(t)$ is measured in kgCO_2 and the cumulative carbon emissions are given by $C_i = \sum_t \Delta C_i(t)$. Let $E_i =$

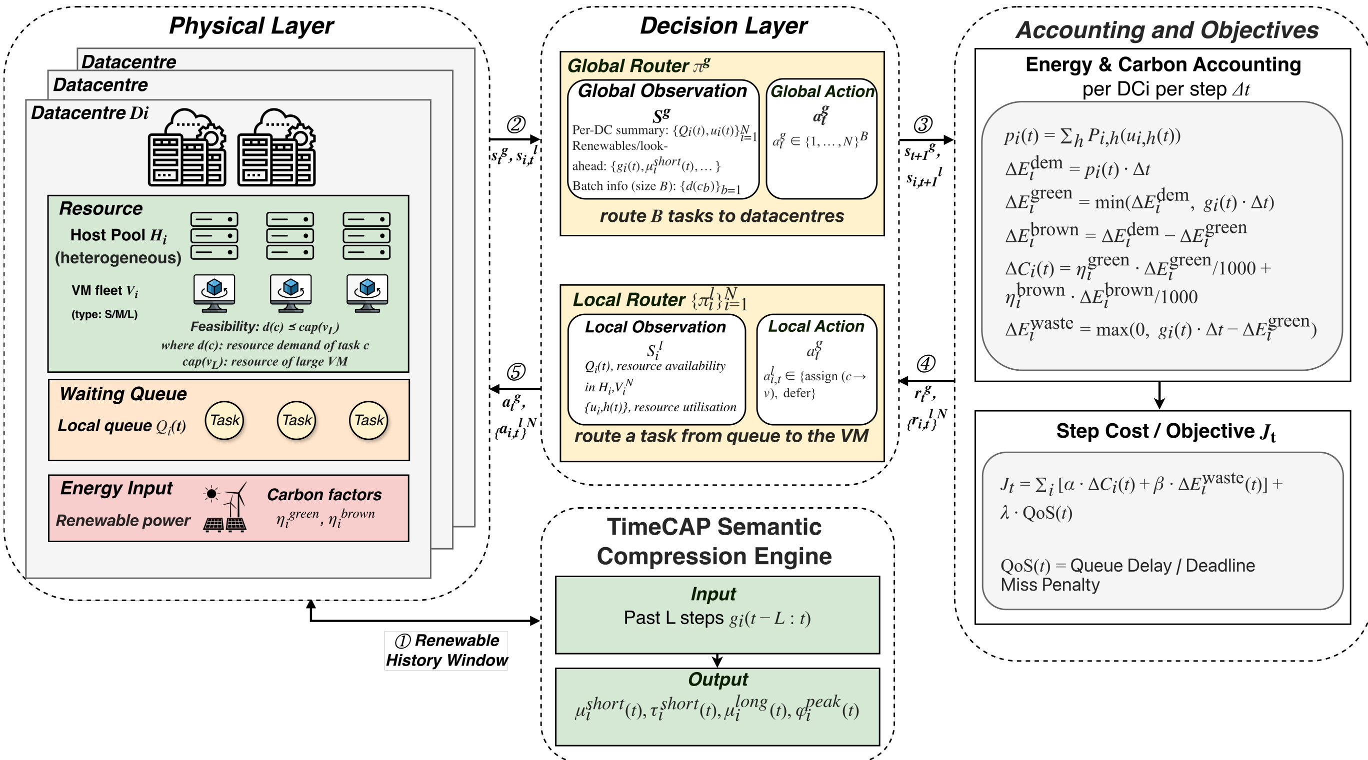

Fig. 1. High-level overview of hierarchical system scheduling architecture in multi-datacentre environments.

$\sum_t \left(\Delta E_i^{\text{green}}(t) + \Delta E_i^{\text{brown}}(t)\right)$ be the total consumed energy (Wh) in an episode; the datacentre carbon intensity is given by:

$$\text{CI}_i = \frac{C_i}{E_i/1000} \qquad (\text{kgCO}_2/\text{kWh}). \tag{9}$$

D. Problem Formulation

Table I summarises the main notation used throughout this work. Given the system model and energy-carbon accounting defined in Sections III-B and III-C, the geo-distributed scheduling problem is formulated as a hierarchical sequential decision-making problem. We formulate the scheduling problem as a spatiotemporally decoupled hierarchical Markov decision process. At each step t, the system state bifurcates into a global routing state s_t^g and localised execution states $s_{i,t}^l$. To circumvent the representational entanglement and noise propagation inherent in raw forecasting trajectories, s_t^g employs semantic state compression. It structurally fuses implicit operational metrics (current datacentre conditions) with explicit low-dimensional structural priors distilled from the TimeCAP backbone. This compact dual-temporal representation equips the global agent with stable, proactive foresight. Concurrently, $s_{i,t}^l$ strictly encapsulates instantaneous node-level conditions (e.g., resource utilisation and queue backlog) to enforce physical feasibility, operating entirely decoupled from global predictive signals. Existing carbon-aware scheduling models are typically reactive or single-layered, treating workload placement as an instantaneous decision based on current system states, and thus fail to capture the temporal foresight and cross-datacentre coordination required under intermittent renewable supply and spatially heterogeneous carbon intensity.

The action space is also hierarchical. At the global level, a routing action $a_t^g = (a_{t,1}, \dots, a_{t,B}) \in \{1, \dots, N\}^B$ assigns a batch of B waiting tasks to target datacentres. At the local level, each datacentre executes a scheduling action $a_{i,t}^l$ that assigns the head-of-line task in its FIFO queue to a feasible VM or defers execution.

State transitions are induced by stochastic task arrivals, task execution dynamics, and time-varying renewable generation. The transition dynamics are not known *a priori* and are implicitly determined by the simulator.

The goal is to learn a hierarchical policy $\pi = (\pi^g, \{\pi_i^l\}_{i=1}^N)$ that minimises the expected discounted cumulative system cost over an episode:

$$\min_{\pi} \ \mathbb{E}\left[\sum_{t=0}^{T} \gamma^t \Big(w_1 \sum_{i=1}^{N} \Delta C_i(t) + w_2 R_w(t) + w_3 \mathcal{L}_{\text{QoS}}(t)\Big)\right], \tag{10}$$

where $\gamma \in (0, 1]$ is the discount factor, and w_1, w_2, w_3 are positive weight parameters that balance the multi-objective trade-offs. The first term penalises system-wide carbon emissions, computed according to the energy and carbon model in Section III-C. The second term penalises renewable energy

TABLE I
NOTATION USED IN THE PROPOSED SYSTEM

Notation	Meaning
N	Number of geo-distributed datacentres
DC_i	Datacentre i
$\mathcal{H}_i$	Host set in DC_i
$\mathcal{V}_i$	VM fleet in DC_i
c	Cloud task (cloudlet)
$d(c)$	Resource demand of task c (e.g., required PEs and MI)
$u_{i,h}(t)$	CPU utilisation of host $h \in \mathcal{H}_i$ at time t
$p_{i,h}(t)$	IT power of host $h \in \mathcal{H}_i$ at time t (W)
$p_i(t)$	IT power of DC_i at time t (W)
$g_i(t)$	Available renewable power at DC_i at time t (W)
Δt	Duration of an accounting step (hours)
$\Delta E_i^{\text{green}}(t)$	Renewable energy used by DC_i at step t (Wh)
$\Delta E_i^{\text{brown}}(t)$	Grid energy used by DC_i at step t (Wh)
$\Delta E_i^{\text{waste}}(t)$	Wasted renewable energy at DC_i (Wh)
$\eta_i^{\text{green}}, \eta_i^{\text{brown}}$	Carbon factors of green and brown energy ($kgCO_2$/kWh)
$\Delta C_i(t)$	Carbon emissions of DC_i at step t ($kgCO_2$)
$Q_i(t)$	Queue length at DC_i
$\overline{T}_i^{\text{wait}}(t)$	Average waiting time at DC_i
s_t^g, a_t^g	Global state and routing action at step t
$s_{i,t}^l, a_{i,t}^l$	Local state and scheduling action at DC_i
π	Hierarchical scheduling policy $\pi = (\pi^g, \{\pi_i^l\})$

curtailment through the renewable waste ratio

$$R_w(t) = \frac{\sum_i \Delta E_i^{\text{waste}}(t)}{\sum_i \left(\Delta E_i^{\text{waste}}(t) + \Delta E_i^{\text{green}}(t)\right)}, \tag{11}$$

and the third term captures QoS degradation:

$$\mathcal{L}_{\text{QoS}}(t) = \frac{1}{N}\sum_{i=1}^{N}\Big(\lambda_1\,\overline{T}_i^{\text{wait}}(t) + \lambda_2\,Q_i(t)\Big). \tag{12}$$

This objective approximates a multi-objective optimisation that jointly minimises carbon emission and renewable energy wastage while constraining excessive queueing delay and backlog. The hierarchical structure enables the decomposition of global routing and local resource allocation decisions, while reinforcement learning is employed to handle stochasticity, partial observability, and long-term trade-offs. Although the objective is defined at the system level, all energy consumption and queueing penalties are induced by individual task executions, thereby implicitly optimising carbon cost and completion efficiency at the per-task level.

IV. HiGreen: Hierarchical Scheduling Algorithm based on Green-Power Look-Ahead

A. Overview and Learning Workflow

This section presents HiGreen, a hierarchical DRL scheduler instantiated via our proposed spatiotemporally decoupled paradigm. To systematically overcome the representational entanglement and the high-frequency noise inherent in raw energy forecasting, HiGreen avoids traditional reactive architectures in favour of an explicit-implicit dual temporal modelling framework.

Decision-making is bifurcated into two specialised tiers. At the macro level, a global routing policy π^g utilises a GTrXL backbone to synthesise implicit historical operational patterns with explicit future foresight, proactively aligning arriving workloads with anticipated renewable surpluses across geo-distributed datacentres. Concurrently, at the micro level, local scheduling policies $\{\pi_i^l\}_{i=1}^N$ strictly operate on instantaneous, node-level capacity constraints to dispatch tasks to feasible VMs via action masking, isolating the physical execution boundaries from global predictive uncertainties. The overall hierarchical routing-and-placement procedure is summarised in Algorithm 2.

At each decision step t, the global agent first invokes TimeCAP to obtain look-ahead features for all datacentres, then collects an aggregated observation s_t^g that combines per-datacentre operational metrics, TimeCAP features, and the current task batch descriptors (Algorithm 1). The global policy π^g outputs a routing action $a_t^g \in \{1,\dots,N\}^B$ that assigns each task in the batch to a target datacentre. Conditioned on this routing, each datacentre i exposes a local observation $s_{i,t}^l$ to its local policy, which then outputs a VM-selection action $a_{i,t}^l$ from a feasibility-constrained action set (implemented via action masking) (Algorithm 2). After executing the global and local actions, the environment advances to the next step and returns the next observations $(s_{t+1}^g, \{s_{i,t+1}^l\})$, together with reward signals reflecting the carbon–waste–QoS trade-off. Transitions collected from the global and local policies are stored in rollout buffers, and PPO updates are performed periodically using the collected trajectories. During training, local policies may share parameters across datacentres to improve sample efficiency, while execution remains decentralised.

B. Renewable Forecasting Backbone: TimeCAP

To provide foresight on renewable availability, the proposed scheduler leverages a TimeCAP-based forecasting backbone [13]. TimeCAP is a channel-aware pre-training framework for multivariate time-series forecasting that emphasises transferable representations by explicitly modelling multivariate dependencies while avoiding unnecessary entanglement with temporal dynamics.

TimeCAP considers a look-back to multi-horizon forecasting formulation. Given a multivariate input series $\mathbf{X} \in \mathbb{R}^{C\times L}$, the objective is to predict the future sequence $\mathbf{Y} \in \mathbb{R}^{C\times H}$, where C denotes the number of channels and L and H denote the input length and forecasting horizon respectively.

Architecturally, TimeCAP integrates reversible instance normalisation (RevIN) to mitigate distribution shift and applies inverse normalisation after decoding to restore the original scale. It then performs multi-scale patch-wise tokenisation to obtain fine-grained temporal tokens. To model multivariate structure with controllable complexity, the channels are partitioned into overlapping groups and embedded in parallel; adaptive meta-routers are introduced to orchestrate cross-group

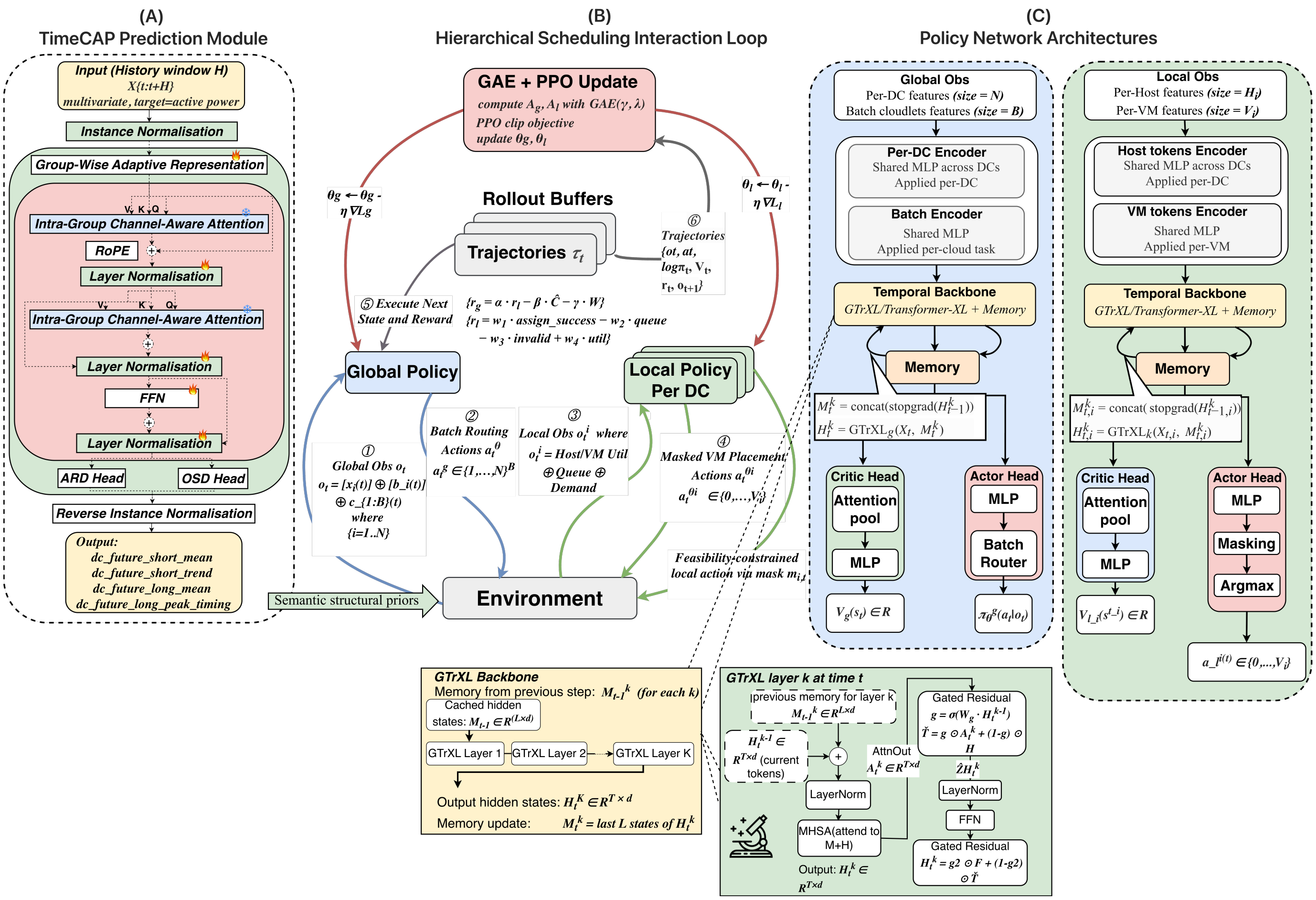

Fig. 2. The proposed HiGreen architecture: An explicit-implicit dual temporal modelling framework integrating the TimeCAP semantic state compression engine with a hierarchical DRL scheduler.

information flow and preserve global coherence. Multivariate dependencies are captured via channel-aware self-attention and cross-attention equipped with channel-aware masks, which confine interactions among time-aligned fine-grained tokens; rotary positional embedding (RoPE) is adopted to encode relative positions for attention.

Crucially, naively exposing the RL scheduling environment to the raw, high-dimensional multi-step predictions $Y \in \mathbb{R}^{C\times H}$ generated by TimeCAP would precipitate a severe state-space explosion, destabilising the policy optimisation process. To achieve a deep, mathematically sound integration between the forecasting backbone and the DRL scheduler, we repurpose TimeCAP as a semantic state compression engine.

Rather than injecting raw trajectories, we distil the volatile future dynamics into a low-dimensional subspace of structural priors. Specifically, we extract the short-term dynamics (mean μ_i^{short} and trend τ_i^{short} over the next k_s intervals) to capture immediate renewable bursts, and long-term structural patterns (mean μ_i^{long} and peak timing ϕ_i^{peak} over k_l intervals) to identify macro-level energy shifts. This explicit temporal representation uniquely equips the global routing policy to anticipate periods of high or low renewable availability, translating raw predictive intelligence into a tractable and highly efficient reinforcement learning state space.

C. State and Action

To operationalise the spatiotemporally decoupled scheduling paradigm, we define a two-level action and observation space. At each decision step t, the global routing policy π^g performs macro-level spatial orchestration for a batch of B waiting tasks across datacentres, while N independent local policies $\{\pi_i^l\}_{i=1}^N$ handle micro-level dispatching to feasible VMs based on instantaneous physical constraints.

Global state. Formulating the global observation s_t^g requires integrating future awareness without inflating the observation space. Rather than ingesting raw prediction sequences, s_t^g leverages abstracted forecasting indicators to maintain representational compactness. Specifically, it systematically synthesises real-time datacentre telemetry with the distilled green-power signatures, alongside the characteristics of the incoming task batch:

$$s_t^g = \Big(\{\mathbf{g}_i(t)\}_{i=1}^N,\ \mathbf{c}^{\text{batch}}(t),\ L_{\text{imb}}(t),\ C_{\text{recent}}(t)\Big). \quad (13)$$

Algorithm 1: HiGreen (Global): Proactive Batch Routing via Dual-Temporal Modelling

Input: Global policy $\pi^g_{\theta_g}$; global batch size B; global queue Q^g; datacentres $\{DC_i\}_{i=1}^N$
Output: Routing action a^g_t and routed batch $\mathcal{B}_t$
Receive arriving tasks $\mathcal{C}_t$; append $\mathcal{C}_t$ to Q^g;
`// Phase 1: Semantic State Compression via TimeCAP Backbone`
foreach *datacentre* DC_i **do**
  Retrieve historical multivariate context $\mathbf{X}_i(t)$;
  `// Distil future volatility into explicit multi-scale structural priors`
  $\left(\mu_i^{\text{short}}, \tau_i^{\text{short}}, \mu_i^{\text{long}}, \phi_i^{\text{peak}}\right) \leftarrow$ TimeCAP_Distil($\mathbf{X}_i(t)$);
`// Phase 2: Dual-Temporal Observation Construction`
Construct global observation s^g_t structurally fusing implicit datacentre aggregates, explicit TimeCAP priors, and batch features;
`// Phase 3: Proactive Spatial Orchestration`
Select batch $\mathcal{B}_t \leftarrow$ PeekBatch(Q^g, B) (pad if $|\mathcal{B}_t| < B$);
Sample routing vector $a^g_t \sim \pi^g_{\theta_g}(\cdot \mid s^g_t)$, where $a^g_t \in \{1, \dots, N\}^B$;
foreach *task* $c_j \in \mathcal{B}_t$ **do**
  Route c_j to target datacentre $DC_{a^g_{t,j}}$;
  Append c_j to the corresponding local FIFO queue $Q_{a^g_{t,j}}$;
Remove $|\mathcal{B}_t|$ tasks from Q^g;

For each datacentre DC_i, the aggregated state vector $\mathbf{g}_i(t)$ is defined as

$$\begin{aligned}\mathbf{g}_i(t) = \Big(&P_i^{\text{green}}(t),\, P_i^{\text{tot}}(t),\, \rho_i^{\text{green}}(t),\, W_i^{\text{green}}(t),\\ &\mu_i^{\text{short}}(t),\, \tau_i^{\text{short}}(t),\, \mu_i^{\text{long}}(t),\, \phi_i^{\text{peak}}(t),\\ &Q_i(t),\, U_i^{\text{cpu}}(t),\, A_i^{\text{pes}}(t),\, U_i^{\text{ram}}(t)\Big).\end{aligned} \tag{14}$$

where, $P_i^{\text{green}}(t)$ is the current available green power (W), $P_i^{\text{tot}}(t)$ is the current IT power draw (W), $\rho_i^{\text{green}}(t) \in [0, 1]$ is the green-energy usage ratio, and $W_i^{\text{green}}(t)$ is the cumulative wasted green energy (Wh).

Crucially, the tuple $\left(\mu_i^{\text{short}}(t), \tau_i^{\text{short}}(t), \mu_i^{\text{long}}(t), \phi_i^{\text{peak}}(t)\right)$ constitutes the explicit multi-scale structural priors distilled from the TimeCAP forecasting backbone. Rather than using reactive states, these specific features represent the short-term mean and trend, alongside the long-term mean and peak timing of renewable availability. By mathematically bounding the non-stationary renewable dynamics into these four semantic indicators, we successfully shield the global routing agent from redundant dimensional complexities while preserving the exact actionable intelligence required for proactive carbon-aware shifting. Finally, $Q_i(t)$ is the local waiting-queue length, $U_i^{\text{cpu}}(t) \in [0, 1]$ and $U_i^{\text{ram}}(t) \in [0, 1]$ denote average host CPU/RAM utilisation, and $A_i^{\text{pes}}(t)$ is the total number of available PEs across active VMs in DC_i.

Algorithm 2: HiGreen (Local): VM Placement Within Each Datacentre

Input: Local policy $\pi^l_{\theta_l}$ (shared or per-DC); local FIFO queues $\{Q_i\}_{i=1}^N$; VM sets $\{\mathcal{V}_i\}_{i=1}^N$
Output: Local placement actions $\{a^l_{i,t}\}_{i=1}^N$
for $i = 1$ **to** N **do**
  if Q_i *is not empty* **then**
    Construct local observation $s^l_{i,t}$ (host/VM utilisation, VM types, free PEs, queue info, next-task demand);
    Compute feasibility mask $m_{i,t} \in \{0,1\}^{|\mathcal{V}_i|+1}$ for the next task in Q_i;
    Sample action $a^l_{i,t} \sim \pi^l_{\theta_l}(\cdot \mid s^l_{i,t}, m_{i,t})$, where $a^l_{i,t} \in \{0, 1, \dots, |\mathcal{V}_i|\}$;
    if $a^l_{i,t} \neq 0$ **then**
      Pop the head task from Q_i; assign it to VM $a^l_{i,t}$ in DC_i;

The batch-level task features are

$$\mathbf{c}^{\text{batch}}(t) = \Big(C^{\text{up}}(t),\; \{p^{\text{req}}_{t,j}\}_{j=1}^B,\; \{m^{\text{req}}_{t,j}\}_{j=1}^B,\; \mathbf{d}^{\text{pes}}(t)\Big), \tag{15}$$

where, $C^{\text{up}}(t)$ is the number of tasks currently waiting in the global queue, $p^{\text{req}}_{t,j}$ and $m^{\text{req}}_{t,j}$ are the PE, and MI demand of the j-th task in the routed batch (padded with zeros if fewer than B tasks are available), and $\mathbf{d}^{\text{pes}}(t) \in \mathbb{N}_0^3$ is the PE-demand distribution of waiting tasks (small/medium/large). We further include cross-datacentre statistics such as utilisation imbalance $L_{\text{imb}}(t)$ (e.g., variance across $\{U_i^{\text{cpu}}(t)\}$) and recent completion count $C_{\text{recent}}(t)$.

Global action. To execute the proactive spatial alignment, the global policy outputs a routing vector

$$a^g_t = (a^g_{t,1}, \dots, a^g_{t,B}) \in \{1, \dots, N\}^B, \tag{16}$$

where, $a^g_{t,j} = k$ assigns the j-th task in the batch to datacentre DC_k. After routing, tasks are appended to the corresponding local FIFO queues.

Local state. To enforce strict physical feasibility entirely decoupled from global predictive uncertainties, the local observation $s^l_{i,t}$ strictly encapsulates instantaneous node-level conditions for each datacentre DC_i:

$$\begin{aligned}s^l_{i,t} = \Big(&\mathbf{h}_i^{\text{load}}(t),\, \mathbf{h}_i^{\text{ram}}(t),\, \mathbf{v}_i^{\text{load}}(t),\, \mathbf{v}_i^{\text{type}}(t),\\ &\mathbf{v}_i^{\text{map}}(t),\, \mathbf{v}_i^{\text{free}}(t),\, Q_i(t),\, p_i^{\text{next}}(t)\Big).\end{aligned} \tag{17}$$

Here, $\mathbf{h}_i^{\text{load}}(t)$ and $\mathbf{h}_i^{\text{ram}}(t)$ are padded vectors of host CPU and RAM utilisation, $\mathbf{v}_i^{\text{load}}(t)$ is the padded VM CPU utilisation vector, $\mathbf{v}_i^{\text{type}}(t) \in \{0, 1, 2, 3\}^{|\mathcal{V}_i|}$ encodes VM types (0=off/empty, 1=small, 2=medium, 3=large), $\mathbf{v}_i^{\text{map}}(t)$ maps each VM slot to its hosting physical machine (or -1 if empty), and $\mathbf{v}_i^{\text{free}}(t)$ gives the number of free PEs per VM. Finally, $Q_i(t)$ is the local queue length and $p_i^{\text{next}}(t)$ is the PE demand of the next task in the scheduling pipeline.

Local action and feasibility masking. To guarantee the physical validity of local placement, the local policy selects a VM (or defers) via

$$a_{i,t}^l \in \{0, 1, \ldots, |\mathcal{V}_i|\}, \tag{18}$$

where action 0 denotes no assignment (keep waiting), and action $k \geq 1$ assigns the next waiting task to VM k. To prevent infeasible assignments during exploration, we employ an action mask $m_{i,t} \in \{0,1\}^{|\mathcal{V}_i|+1}$, where $m_{i,t}(k) = 1$ if VM k has sufficient available resources for the next task (and $m_{i,t}(0) = 1$ always holds); masked actions are strictly disallowed by the policy to preserve execution boundaries.

D. Reward Design for Carbon Optimisation

Our reward design intends to minimise carbon emissions while trading off carbon reduction against QoS. To this end, HiGreen adopts a hierarchical reward structure in which the global and local agents receive different but complementary signals and cooperate to achieve system-wide carbon efficiency without sacrificing service performance. Local policies are encouraged to reduce waiting time, improve VM utilisation, and avoid infeasible actions, while the global policy additionally penalises carbon emissions and renewable curtailment.

Local reward. For each datacentre DC_i, we define the local reward at step t as

$$r_{i,t}^l = r_{i,t}^{\text{wait}} + r_{i,t}^{\text{util}} + r_{i,t}^{\text{queue}} + r_{i,t}^{\text{inv}}, \tag{19}$$

where the components follow the simulator implementation. First, the waiting-time penalty is computed from the set of waiting times of tasks that finish within the current step:

$$r_{i,t}^{\text{wait}} = -\omega_{\text{wait}} \log\big(1 + \overline{T}_{i,t}^{\text{wait}}\big). \tag{20}$$

Second, we penalise low utilisation and imbalance using active VM CPU utilisation in DC_i. Let $\bar{U}_{i,t}^{\text{vm}}$ be the mean VM utilisation and $\sigma_{i,t}^{2,\text{vm}}$ be the corresponding variance; then

$$r_{i,t}^{\text{util}} = -\omega_{\text{util}} \Big(\sqrt{\sigma_{i,t}^{2,\text{vm}}} + \big|\bar{U}_{i,t}^{\text{vm}} - U^\star\big| \Big), \tag{21}$$

where, $U^\star$ is a target utilisation (set to 0.75 in our multi-datacentre setting). Third, we penalise queue backlog by normalising the local queue length $Q_i(t)$ by the cumulative number of tasks routed to DC_i, denoted by $N_i^{\text{recv}}(t)$:

$$r_{i,t}^{\text{queue}} = -\omega_{\text{queue}} \frac{Q_i(t)}{\max\{1,\, N_i^{\text{recv}}(t)\}}. \tag{22}$$

Finally, infeasible actions are penalised with a flat term:

$$r_{i,t}^{\text{inv}} = -\omega_{\text{inv}} \cdot \mathbb{I}_{i,t}^{\text{inv}}, \tag{23}$$

where $\mathbb{I}_{i,t}^{\text{inv}} = 1$ if the raw neural network output attempts to select an invalid action prior to masking. Although the environment strictly rejects masked actions to preserve physical safety, applying this penalty to the raw policy logits significantly accelerates the network's learning of the feasibility boundaries. (e.g., selecting a non-existent VM or attempting placement with an empty queue), and 0 otherwise.

Global reward for carbon optimisation. The global reward combines average local performance with system-wide carbon and renewable-waste penalties:

$$r_t^g = \alpha\, \bar{r}_t^l - \beta\, \hat{C}_t - \kappa\, R_w(t), \tag{24}$$

where, $\bar{r}_t^l = \frac{1}{N}\sum_{i=1}^N r_{i,t}^l$. The carbon penalty uses a running-max normalisation to stabilise learning:

$$\hat{C}_t = \min\left(\frac{\sum_{i=1}^N \Delta C_i(t)}{M_C + \epsilon},\, c_{\max} \right), \tag{25}$$

where, M_C is the running maximum of $\sum_i \Delta C_i(t)$ observed so far, ϵ avoids division by zero, and $c_{\max}$ caps extreme ratios. The renewable-waste ratio is

$$R_w(t) = \frac{\sum_{i=1}^N \Delta E_i^{\text{waste}}(t)}{\sum_{i=1}^N \big(\Delta E_i^{\text{waste}}(t) + \Delta E_i^{\text{green}}(t)\big)} \in [0,1]. \tag{26}$$

This design encourages the global policy to route workloads towards datacentres with adequate renewable energy availability, while local policies maintain QoS and feasibility.

E. Policy Network Architecture and Parameter Sharing

We parameterise both the global routing policy π^g and the local scheduling policies $\{\pi_i^l\}_{i=1}^N$ with neural networks that map observations to action distributions. To capture temporal dependencies induced by non-stationary workloads and time-varying renewable supply, we adopt GTrXL as the policy backbone. While standard Transformer architectures are notoriously unstable in deep reinforcement learning, GTrXL overcomes this by incorporating a gated residual network. This gating mechanism dynamically modulates the information flow between layers, acting as a robust identity map that stabilises policy optimisation without sacrificing the representational power of the temporal memory. Unless otherwise stated, the global and local policies use the same backbone configuration while maintaining separate heads for their respective action spaces.

At each step, the global network receives the aggregated observation s_t^g, which includes per-datacentre metrics, Time-CAP foresight features, and batch-level task descriptors. The local network receives $s_{i,t}^l$, comprising host/VM utilisation vectors, VM-type indicators, resource availability, and queue-related features. Continuous features are normalised to $[0,1]$ where appropriate, and categorical features (e.g., VM types) are represented via learned embedding vectors. The resulting feature tokens are then fed into the GTrXL layers.

GTrXL employs segment-level recurrence to maintain a fixed-length memory of past hidden states, enabling the policy to leverage historical context beyond a single step. In our implementation, the backbone is configured with L Transformer layers, hidden dimension d, H attention heads, feed-forward width d_{ff}, and memory length m, as specified in the experimental configuration. This design allows the agent to infer temporal patterns such as renewable trends and queue build-up without explicitly hand-crafting history features. The combination of GTrXL's temporal modelling with TimeCAP's explicit look-ahead features provides complementary sources of temporal information. GTrXL captures implicit patterns from observation history, while TimeCAP supplies explicit forecasts of future renewable availability.

The global policy head outputs a routing distribution over $\{1, \dots, N\}^B$ for batch routing, while each local policy head outputs a distribution over $\{0, 1, \dots, |\mathcal{V}_i|\}$, where action 0 denotes deferring placement. To ensure feasibility, we apply action masking to the local head by zeroing out logits of invalid VM choices (e.g., VMs with insufficient free PEs for the next task) and renormalising the remaining probabilities.

To improve sample efficiency and reduce the number of trainable parameters, we adopt a CTDE paradigm with parameter sharing among local schedulers. All local agents share the same network parameters θ_l while operating on different local observations. Training is centralised in the sense that trajectories from both the global policy and all local policies are collected and optimised jointly via PPO. At runtime, execution remains fully decentralised: each agent acts solely based on its own observation without requiring communication with other agents, thereby enabling scalable deployment across geographically distributed datacentres.

V. Performance Evaluation

A. Experiment Setup

This work is tested by utilising the CloudSim Plus framework [39], a discrete-event simulator for cloud datacentre environments. CloudSim Plus provides the core simulation infrastructure in our implementation, including cloud task execution, VM provisioning, host resource accounting, and power modelling. We model geo-distributed datacentres with heterogeneous compute infrastructures. Each datacentre is configured with a single server profile (host type) and a pre-provisioned VM fleet with three VM sizes (Small/Medium/Large). Host power consumption is modelled as a function of CPU utilisation using empirical server power curves derived from the SPECpower_ssj2008 benchmark suite [40], and the adopted server profiles are summarised in Table II. These curves provide realistic idle-to-peak power characteristics for different server platforms, and are used by CloudSim Plus to compute instantaneous IT power based on utilisation. For the DRL framework and environment interface, the learning pipeline Ray RLlib [41] is applied, and the Java-based simulator is exposed by Py4J and PettingZoo [42] as a multi-agent environment. The hierarchical setting is represented as a global routing agent and multiple local scheduling agents (one per datacentre), enabling centralised training and decentralised execution within RLlib. Each datacentre d is associated with a renewable generation trace (wind or solar) extracted from public datasets [43], [44], which provides time-varying green power availability during simulation. While our full evaluation environment comprises ten geo-distributed datacentres, Fig. 3 illustrates the representative daily variations of green power for a subset of five datacentres to maintain visual clarity.

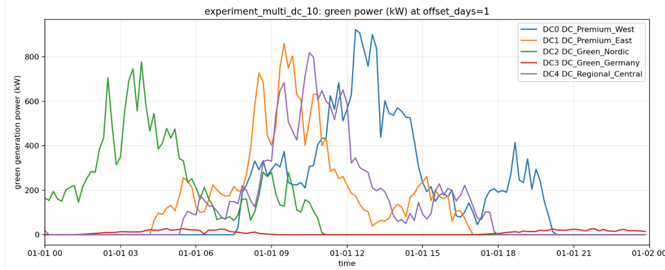

Fig. 3. Daily Green Energy Production Variation of 5 Datacentres

B. Workload Dataset

We evaluate HiGreen under three synthetic workload intensities (comprising 5k, 15k, and 100k cloudlets respectively) generated by a custom stochastic workload profiler. To rigorously reflect the highly heterogeneous nature of modern cloud computing environments, each atomic cloud task (cloudlet) is mathematically profiled by a multivariate tuple comprising its arrival timestamp, execution length (Million Instructions, MI), parallel processing element (PE) demand, and data payload size.

Arrival Dynamics. Workload arrivals are modelled as a stochastic point process. Unless otherwise specified, task arrivals follow a Poisson process with an arrival rate λ (yielding exponential inter-arrival times) over an evaluation duration T. The underlying generator is also capable of simulating uniform and bursty temporal distributions to evaluate system robustness under extreme traffic spikes.

Task Characteristics. The computational footprint of each task is sampled from a bounded distribution within $[L_{\min}, L_{\max}]$ measured in MI. To accurately emulate real-world datacentre workload distributions, the PE demand is drawn from a discrete set $\{1, \dots, 8\}$ following a right-skewed probability mass function, ensuring a realistic predominance of smaller jobs. Furthermore, the I/O payload (input and output data sizes) is explicitly defined as a linear function of the execution length (approximately 1 KB per 1000 MI).

Evaluation Consistency. To ensure rigorous and reproducible benchmarking, all stochastic workload traces are generated using fixed pseudo-random seeds. Crucially, the 5k, 15k, and 100k workload scenarios scale exclusively in terms of temporal arrival intensity λ, while the underlying statistical distributions of task sizes and PE demands remain strictly invariant. This isolates the independent variable, allowing for a controlled and scientifically sound evaluation of algorithmic scalability.

C. Baselines

We compare against seven representative baselines. Round-Robin is a simple heuristic that routes incoming tasks to datacentres in a cyclic order, serving as a lightweight load-balancing reference. GA (Genetic Algorithm) and PSO (Particle Swarm Optimisation) are population-based metaheuristics that search for routing/scheduling decisions by iteratively improving candidate solutions via evolutionary operators (GA) or swarm updates (PSO), providing strong optimisation-oriented baselines. PPO-Simple (no prediction) is a reinforcement-learning baseline trained with Proximal Policy Optimisation using only current observable system states (without renewable prediction features), capturing the benefit of RL without foresight. Under the oracle-style global observation ("god-eye") setting with prediction features, we include three policy-network baselines: PPO-MLP, PPO-gMLP [45], and PPO-ResMLP [46], which share the same training algorithm and observation access but differ in function approximation capacity/inductive bias. Finally, our proposed method is HiGreen, which employs a GTrXL-style policy under the aforementioned explicitly formulated observation space to capture deep temporal dependencies. Collectively, these baselines effectively represent the traditional scheduling paradigms. Specifically, heuristic methods (e.g., GA and PSO) and the standard predictive-free PPO strictly operate on a reactive, instantaneous-state basis. By benchmarking HiGreen against these baselines, we move beyond merely comparing scalar rewards to systematically validating the scientific necessity of our proposed spatiotemporal decoupling and demonstrating the quantifiable carbon-reduction benefits yielded by the explicit-implicit dual temporal foresight mechanism.

D. Results

This work systematically evaluates HiGreen through training dynamics analysis and comparative benchmarking. Fig. 4 illustrates the training reward evolution. The synchronous ascent of the global routing reward and average local placement reward demonstrates a robust, stable co-evolution. This confirms that our explicit-implicit formulation effectively mitigates multi-agent non-stationarity and credit assignment challenges, aligning both scheduling tiers towards a unified carbon-reduction objective.

Fig. 5 dissects the individual local agents. Despite extreme spatial heterogeneity in datacentre capacities and renewable availability, all ten agents converge robustly. Although initiating from disparate baselines (ranging from $-5,000$ to $-3,200$), their rewards ultimately narrow into a cohesive band, validating that the shared-parameter local policies effectively generalise across diverse physical boundaries.

Fig. 6 captures the environmental impact during training. The progressive reduction in total carbon emissions validates the global agent's proactive divestment from carbon-intensive grid energy. Furthermore, the rapid stabilisation of carbon intensity (around 0.2624 kg/kWh) implies these reductions stem from structural scheduling foresight rather than erratic workload shifting, ensuring operational stability for QoS.

Fig. 7 compares HiGreen against baselines across three workload scales. At the 5k scale, all methods perform comparably due to the lightly constrained environment. However, under the 15k and 100k workloads, reactive baselines (e.g., GA, PSO, PPO-Simple) suffer from short-sighted local optima. In contrast, HiGreen maintains consistent carbon efficiency. This advantage is strictly enabled by the explicit-implicit dual-temporal architecture. By extracting semantic priors from raw green-power signals via TimeCAP, the global agent proactively anticipates spatial energy shifts. Furthermore, while the predictive baselines (PPO-MLP, PPO-gMLP, PPO-ResMLP) have equal access to these TimeCAP priors, they lack the segment-level recurrence required to capture implicit historical trends, such as workload non-stationarity and prolonged queue build-ups. HiGreen's GTrXL backbone combines explicit future priors with implicit historical memory, yielding superior and far more stable long-term coordination, while the decoupled local policies ensure physical feasibility at the node level.

VI. Conclusion

This paper addressed the fundamental spatiotemporal mismatch between volatile renewable energy supply and strict datacentre execution constraints by proposing a novel bilateral green-energy alignment paradigm. To proactively synchronise computing demand with forthcoming renewable availability, we introduced HiGreen, a spatiotemporally decoupled hierarchical DRL framework. Rather than relying on traditional reactive strategies or exposing the learning agent to raw, high-dimensional prediction sequences, HiGreen features an explicit-implicit dual temporal modelling architecture. By leveraging semantic state compression via the TimeCAP forecasting backbone, complex future renewable dynamics are distilled into compact structural priors. The resulting two-level policy decouples forward-looking, cross-datacentre routing from feasibility-constrained, intra-datacentre VM placement. Experimental evaluations under dynamic workload scenarios empirically validate that HiGreen effectively circumvents the representational entanglement inherent in raw predictive DRL, proactively steering workloads towards renewable-rich locations. This significantly improves system-wide carbon efficiency and reduces carbon intensity without compromising practical QoS guarantees. Future work will focus on enhancing the global policy's robustness to inherent forecasting uncertainties, incorporating network-topology constraints, and scaling the framework to more expansive, highly heterogeneous geo-distributed cloud environments.

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TABLE II
SERVER PROFILES USED IN OUR SIMULATOR (DERIVED FROM SPECPOWER_SSJ2008)

Server Name	Processor	Cores	MIPS	RAM	Power (W)	
					Idle	Active
Acer Altos R520	Intel Xeon E5450	8	400k	16 GB	155.0	269.0
HP ProLiant DL360 G7	Intel Xeon 5600 series	8	400k	32 GB	58.0	208.0
Acer AR360 F2	Intel Xeon E5-2660	16	800k	32 GB	69.4	315.0
Dell PowerEdge R720	Intel Xeon E5-2600 series (v1/v2)	16	800k	64 GB	98.0	345.0
Cisco UCS C240 M4	Intel Xeon E5-2600 series (v3/v4)	24	1200k	128 GB	142.0	476.0
HPE ProLiant DL380 Gen10	Intel Xeon Scalable (1st/2nd Gen)	32	1600k	256 GB	194.0	634.0
ASUSTeK RS720-E9/RS8	Intel Xeon Platinum 8180 (2×28)	56	2800k	128 GB	48.2	385.0
ASUSTeK RS500A-E10-PS4	AMD EPYC 7742	64	3200k	256 GB	51.4	214.0
ASUSTeK RS700A-E9-RS4V2	AMD EPYC 7742 (2×64)	128	6400k	512 GB	106.0	430.0

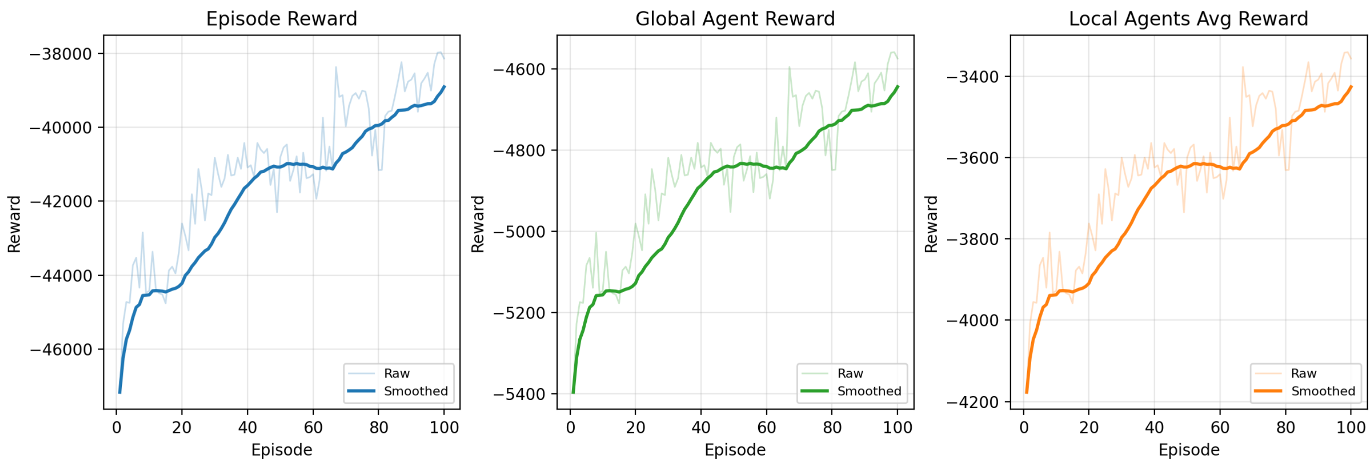

Fig. 4. Training reward curves showing episode reward (left), global agent reward (middle), and average local agent reward (right). All metrics exhibit consistent improvement over 100 training episodes.

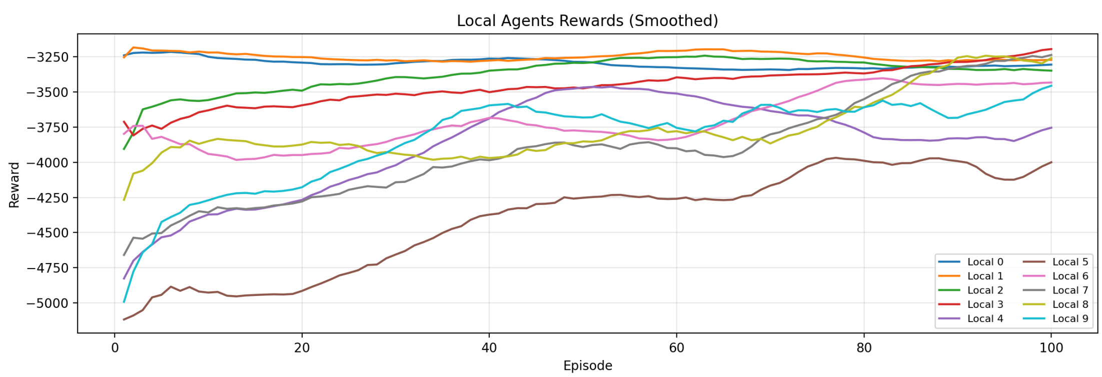

Fig. 5. Individual local agent reward trajectories across ten datacentres. Despite heterogeneous conditions, all agents demonstrate upward learning trends.

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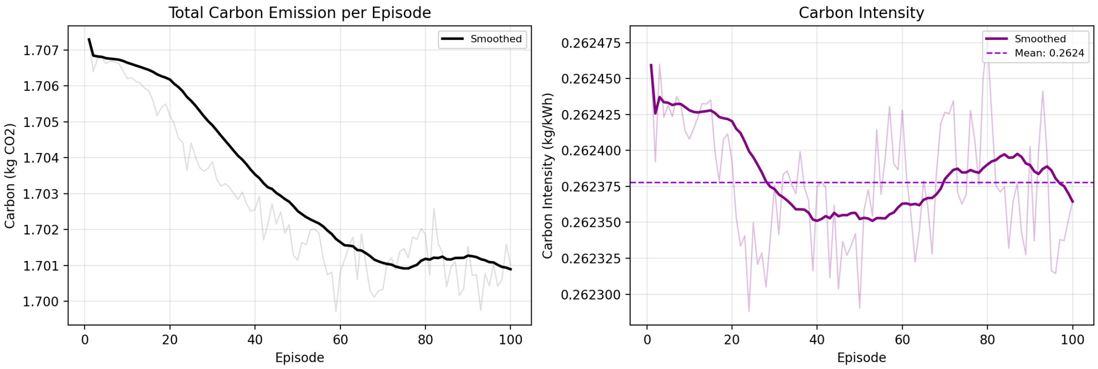

Fig. 6. Carbon metrics during training: total carbon emission per episode (left) and carbon intensity (right). The dashed line indicates the mean carbon intensity of 0.2624 kg/kWh.

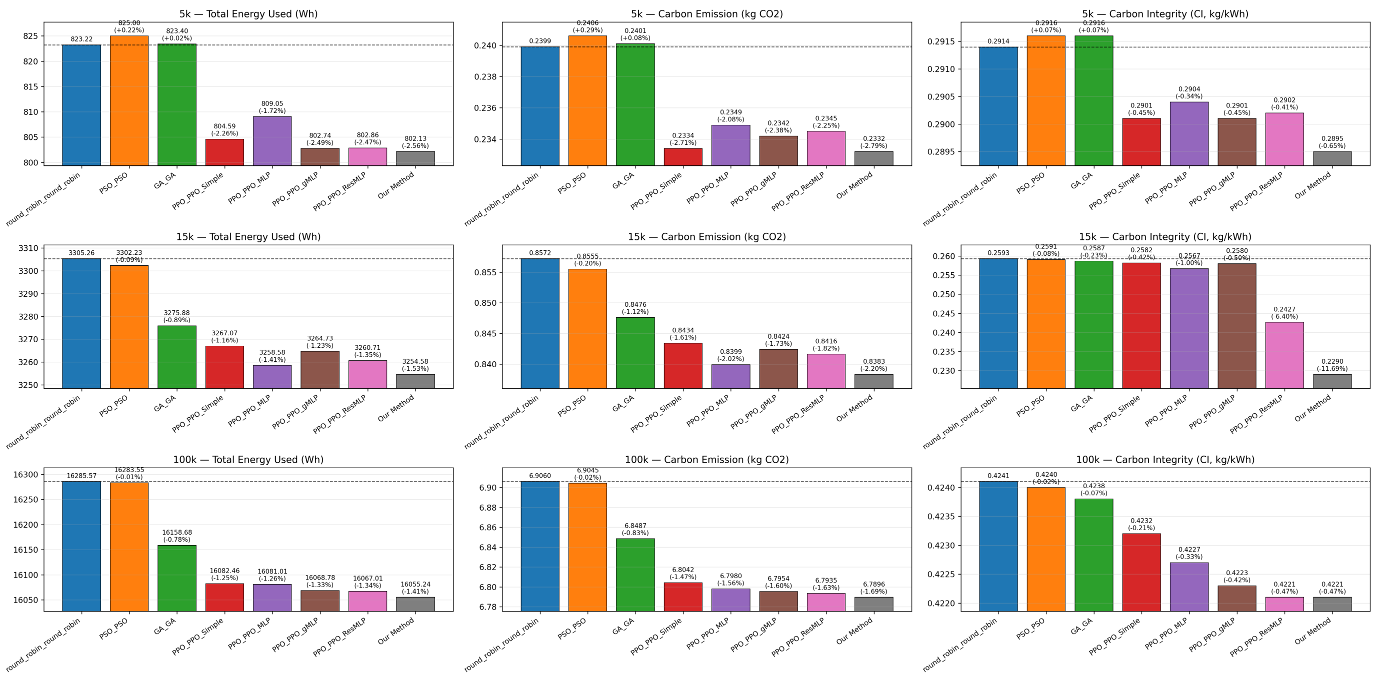

Fig. 7. Performance comparison across different workload scales (5k, 15k, 100k cloudlets) in terms of total energy consumption, carbon emission, and carbon intensity.

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